THE COTANGENT COMPLEX OF A TOR INDEPENDENT TENSOR PRODUCT OF ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays both the decomposition and its representing complex; the proof handles the vanishing and decomposition first and then the representation.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- Flat base change for the cotangent complex. Lemma lemma-flat-base-change-cotangent-complex (1201-1209, produced as L44156), used at proof lines 3051-3053. Gives the vanishing of E and the decomposition.
- The cotangent complex of an algebra is a complex of free modules in degree zero. used at proof lines 3054-3058. Turns the derived tensor product into an ordinary tensor product.
- The definition of the object E in the tensor product equation. equation-tensor-product, used at proof lines 3039-3040. Fixes the object whose vanishing is asserted.

Source reference. Original equation 09D9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11857–11857. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11068–11068. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *If A and B are Tor independent R-algebras, then the object E in (09D9) is zero. In this case we have*

$$L_{A\otimes_R B/R} = L_{A/R} \otimes^{\mathbf{L}}_A (A \otimes_R B) \oplus L_{B/R} \otimes^{\mathbf{L}}_B (A \otimes_R B)$$

which is represented by the complex $L_{A/R} \otimes_R B \oplus L_{B/R} \otimes_R A$ of $A \otimes_R B$-modules.

Proof. The first two statements are immediate from Lemma 08QQ. The last statement follows as $L_{A/R}$ is a complex of free A-modules, hence $L_{A/R} \otimes^{\mathbf{L}}_A (A \otimes_R B)$ is represented by $L_{A/R} \otimes_A (A \otimes_R B) = L_{A/R} \otimes_R B$ □

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